AIS – AURELIA INDUSTRIAL SYSTEMS

AIS-500 Technical Specifications
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1. Document Purpose

This document defines the technical specifications and engineering characteristics of the AIS-500 High Performance Production System.

It provides technical reference information for installation, production integration, diagnostics, maintenance, communication, and technical support.

The document is intended for customers, system integrators, maintenance personnel, technical support engineers, and AIS Engineering.

This document complements the AIS-500 User Manual and does not replace applicable safety documentation.

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2. System Overview

The AIS-500 is a high-throughput automated production system designed for demanding industrial manufacturing environments.

The system integrates:

- multiple robotic production modules;
- automated material handling;
- industrial PLC control;

- advanced diagnostics;
- high-performance production sequencing;
- MES connectivity;
- industrial communication;
- redundant safety architecture;
- monitored safety zones.

The AIS-500 is designed for applications requiring high production throughput and integration with broader manufacturing execution systems.

2.1 Product Identification

Parameter	Specification
Product	AIS-500
Machine type	High Performance Production System
Nominal maximum capacity	180 units/min
Control system	AIS Control PLC-500
HMI	AIS Advanced Industrial HMI
Primary communication	Industrial Ethernet
Additional communication	OPC UA
Enterprise integration	MES integration
Safety architecture	Redundant safety architecture
Preventive maintenance interval	Every 2,500 operating hours

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3. Mechanical Specifications

3.1 Dimensions

The standard AIS-500 configuration has the following approximate dimensions:

Parameter	Value
Length	5,200 mm
Width	2,600 mm
Height	2,800 mm
Approximate weight	8,500 kg

Actual dimensions may vary depending on production modules, tooling, customer configuration, and optional equipment.

The installation area must provide sufficient space for operation, maintenance, material flow, emergency access, and equipment servicing.

3.2 Mechanical Configuration

The AIS-500 consists of a modular high-performance production architecture including:

- primary machine structure;
- multiple robotic modules;
- automated material handling;
- production stations;
- electrical control cabinets;
- industrial PLC;
- operator interface;
- safety zones;
- communication interfaces.

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4. Electrical Specifications

The standard electrical configuration is:

Parameter	Specification
Supply voltage	400 VAC

Supply type	Three-phase
Frequency	50/60 Hz
Control voltage	24 VDC
Main control system	AIS Control PLC-500

The required electrical infrastructure must be verified during installation planning.

Electrical installation, commissioning, and maintenance must be performed by qualified personnel.

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5. Performance and Capacity

The AIS-500 has a nominal maximum production capacity of:

180 units per minute

Actual production performance depends on:

- product characteristics;
- production recipe;
- tooling;
- robotic cycle time;
- material availability;
- production configuration;
- upstream equipment;
- downstream equipment;
- MES integration;
- environmental conditions.

The nominal capacity is a technical maximum under defined operating conditions and should not be interpreted as a guaranteed output for every production application.

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6. Environmental Requirements

The standard operating environment is:

Parameter	Requirement
Operating temperature	5–35 °C
Relative humidity	Up to 80%, non-condensing
Installation environment	Indoor industrial environment
Altitude	Up to 1,000 m without special configuration

Environmental conditions outside the specified operating range may require an approved engineering configuration.

The machine should be protected from excessive dust, corrosive substances, uncontrolled condensation, and other environmental conditions that could affect machine operation.

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7. Control System

The AIS-500 uses the **AIS Control PLC-500** as its primary control system.

The PLC manages:

- production sequencing;
- robotic coordination;
- machine states;
- sensor monitoring;
- actuator control;
- alarm management;
- HMI communication;
- diagnostic information;
- communication with external systems.

The control architecture is designed to support high-throughput production and integration with external manufacturing systems.

PLC software and protected machine parameters must not be modified without appropriate authorization.

8. Safety System

The AIS-500 incorporates a redundant safety architecture with monitored safety zones.

Safety functions include:

- emergency stop circuits;
- guard monitoring;
- safety sensors;
- monitored access zones;
- redundant safety control;
- safe machine-state handling.

Access to defined safety zones may trigger controlled machine stopping or other configured safety responses.

Safety devices must not be bypassed, disabled, or modified.

Any change to safety configuration requires approved engineering procedures and qualified personnel.

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9. Communication Interfaces

The standard communication architecture includes:

- Industrial Ethernet;
- OPC UA;
- MES integration.

9.1 Industrial Ethernet

Industrial Ethernet is used for machine-level network communication and integration with compatible industrial equipment.

9.2 OPC UA

OPC UA may be used to expose selected machine information to compatible supervisory or integration systems.

9.3 MES Integration

The AIS-500 supports integration with Manufacturing Execution Systems.

MES integration may be used for:

- production data exchange;
- production status;
- recipe-related information;
- production monitoring;
- manufacturing traceability.

The exact data exchanged depends on the customer integration project.

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10. HMI and Diagnostics

The AIS Advanced Industrial HMI provides access to:

- machine status;
- production status;
- alarms;
- diagnostic information;
- maintenance information;
- operating parameters;
- selected production information.

The diagnostic system supports technical personnel during troubleshooting and maintenance.

Diagnostic information should be evaluated together with machine history and relevant technical documentation.

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11. Manufacturing System Integration

The AIS-500 is designed to operate as part of a broader manufacturing environment.

Integration may include:

- MES platforms;

- production monitoring systems;
- industrial networks;
- upstream production equipment;
- downstream production equipment.

Integration interfaces must be configured according to the approved customer project.

External system failures may affect machine operation even when the AIS-500 itself is operating correctly.

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12. Utilities and Auxiliary Requirements

The primary utility requirement is the electrical supply specified in this document.

Additional utility requirements depend on the installed configuration and customer production environment.

Where auxiliary systems are required, their specifications are defined during the engineering and installation phases.

Customers must verify all project-specific utility requirements before commissioning.

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13. Maintenance Requirements

Preventive maintenance for the AIS-500 is scheduled every:

2,500 operating hours

Maintenance activities may include:

- robotic system inspection;
- mechanical inspection;
- sensor inspection;
- safety-system verification;
- electrical inspection;
- actuator inspection;
- cleaning;

- communication-system verification;
- functional testing.

Maintenance intervals must be evaluated using the current AIS maintenance documentation and the machine's actual operating hours.

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14. Installation Considerations

Installation of the AIS-500 requires careful planning due to the machine's dimensions and weight.

Installation planning must consider:

- floor load capacity;
- transportation route;
- lifting and positioning equipment;
- electrical infrastructure;
- grounding;
- network infrastructure;
- maintenance clearance;
- safety-zone layout;
- upstream and downstream interfaces;
- MES connectivity.

Final installation requirements are configuration-specific.

Commissioning must be performed according to the approved AIS commissioning procedure.

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15. Configuration and Variants

The AIS-500 supports different production configurations.

Possible configuration differences include:

- number and type of robotic modules;

- production tooling;
- material handling systems;
- safety-zone configuration;
- HMI configuration;
- network configuration;
- MES integration;
- production recipes;
- diagnostic configuration.

Customer-specific engineering documentation takes precedence over generic technical specifications.

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16. Technical Support Data

When contacting AIS Technical Support, provide:

- machine model;
- machine serial number;
- current operating hours;
- production state;
- alarm or error code;
- description of the issue;
- production impact;
- recent maintenance activity;
- relevant diagnostic information;
- MES or communication status where applicable.

For communication-related issues, provide the affected interface and the approximate time at which communication was lost.

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17. Document Control

Version	Date	Description
2.0	2025-01-15	Technical specification revision
2.5	2025-07-10	Updated integration and safety information
3.0	2026-01-20	Current technical specification

Current approved version: 3.0